

the data to the midpoint–time plane. Migration (Section 3.3.7) in the midpoint–time plane seeks to eliminate artifacts due to diffractions and convert the seismic section to an image of the subsurface. The migrated section can then be converted, using assumptions about velocities, to a depth section. The depth section is then interpreted together with geological data and other types of geophysical data, in some cases from drill holes, to understand the subsurface geology.

This discussion of the processing sequence brings out the point that although it is natural to treat a seismic section as an accurate image of the subsurface, it is actually a display of a seismic wave field showing the energy arriving as a function of two-way travel time. Thus the quantity shown, vertical displacement or pressure, need not correspond to any geological reflector of interest. Large arrivals can result from interference between reflections from small impedance contrasts. Moreover, because a seismic section has been produced by mathematical operations, rather than the physical experiment it simulates, noise in the data and errors in the processing can produce spurious artifacts. For example, the conversion of time to depth is only as accurate as the velocities found by stacking or otherwise, perhaps from measurements in a drill hole.

As we have seen in discussing migration, seismic sections are most likely to deviate from the desired images when the medium has significant lateral variations. For example, a medium with random heterogeneities can yield spurious short layered segments, because the reflected energy depends on the vertical changes in impedance. Thus long-wavelength vertical variations in impedance are suppressed, whereas both short- and long-wavelength horizontal variations are preserved, and so can yield a structure with apparent horizontal layering. This effect can be viewed as a velocity filter (Section 3.3.5) that reduces horizontal resolution for structures with steep dips. Similar effects, which are prone to occur at large offsets, may contribute to the horizontally discontinuous layering observed in deep crustal reflection data (Section 3.2.4). Hence, as we will see in various contexts throughout our discussions (e.g., Section 7.3), studying three-dimensional velocity structure is an interesting and challenging enterprise.

3.4 Seismic waves in a spherical earth

In the previous sections, we developed the theory to use the travel times of seismic waves to study the velocity structure of a medium composed of flat layers. This analysis is useful when the ray paths between the source and the receiver are short enough that the earth's curvature can be neglected. Because this is the case for distances less than a few hundred kilometers, such analysis is used to study structure in the crust and the uppermost mantle. In this section we develop the corresponding theory for a spherical earth, which can be used for greater distances and thus greater depths. Application of these results, discussed in the next section, is our primary tool for studying the structure of the deep earth.

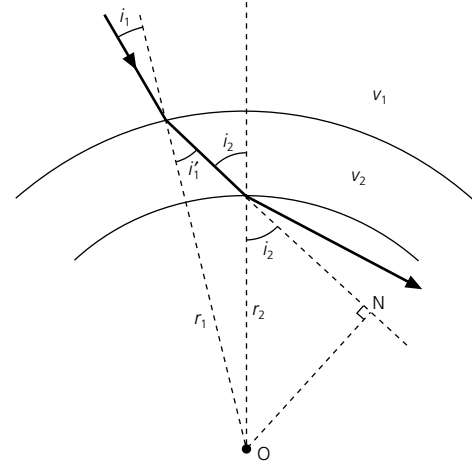


Fig. 3.4-1 Geometry of Snell's law for a spherical earth.

3.4.1 Ray paths and travel times

By analogy to the way we previously represented the earth using uniform flat layers, we now treat it as a series of concentric spherical shells of uniform-velocity material. The ray paths and travel times for the spherical geometry are described by expressions similar to those for flat layers (Section 3.3.1). Consider the portion of a seismic ray's path connecting points at radial distances r_1 and r_2 from the earth's center (Fig. 3.4-1). If v_1 and v_2 are the velocities above and below r_1 , and i_1 , i_1' and i_2 are the angles shown, then by Snell's law

$$\frac{r_1 \sin i_1}{v_1} = \frac{r_1 \sin i_1'}{v_2}. \quad (1)$$

However, $r_1 \sin i_1' = r_2 \sin i_2$ because both equal the length ON , so we rewrite Eqn 1 as

$$\frac{r_1 \sin i_1}{v_1} = \frac{r_2 \sin i_2}{v_2}. \quad (2)$$

Thus we define the ray parameter p for a spherical earth as

$$p = \frac{r \sin i}{v}, \quad (3)$$

where r is the radial distance from the center of the earth, v is the velocity at that point, and i is the incidence angle between the ray path and the radius vector. By reducing the thickness of the shells ever thinner, the velocity becomes a continuous function of radius, $v(r)$. Equation 3 is thus Snell's law for a spherical earth, which describes the ray path. As for the flat earth, the ray parameter is constant along the ray path, and thus identifies a particular ray.

It may seem strange that different forms of the ray parameter and Snell's law occur for a sphere. At any given depth, the flat

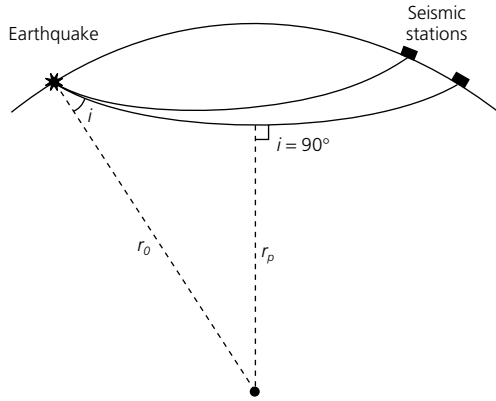


Fig. 3.4-2 Geometry of a ray path in a spherical earth with velocity increasing with depth. The angle of incidence, i , is 90° at the bottoming depth r_p .

layer formulation, that $p = \sin i/v$ is constant, is valid. The factor r corrects for the change along the path of the orientation of the normal to the interface, which is the radius. If r changes so slowly along the path that its variation can be ignored, we obtain the flat case. Thus the flat layer version is used for near-surface refraction and reflection studies.

The condition of constant ray parameter relates the ray path to the velocity structure. For a source at a radius r_0 (the earth's radius for a surface source) where the velocity is v_0 ,

$$p = r_0 \sin i / v_0. \quad (4)$$

Rays leaving the source at different angles thus have different ray parameters. As the ray travels downward, r decreases, and in general v increases, so $\sin i$ and thus i increase, because p is constant. The ray eventually “bottoms” and turns upward when $i = 90^\circ$ (Fig. 3.4-2). At this bottoming depth, $r = r_p$, and

$$p = r_p / v_p. \quad (5)$$

From this point the ray returns to the surface. Different rays, with different p , thus bottom at different depths.

Consider two rays with ray parameters p and $p + dp$, that arrive at nearby points on the earth's surface (Fig. 3.4-3). The ray with ray parameter p takes a travel time T to travel a distance Δ , measured by the angle subtended at the earth's center, whereas the ray with $p + dp$ takes $T + dT$ to travel $\Delta + d\Delta$. In the limit, as the distance between the two points goes to zero,

$$\frac{v_0 dT}{r_0 d\Delta} = \sin i, \quad (6)$$

so

$$\frac{dT}{d\Delta} = \frac{r_0 \sin i}{v_0} = p. \quad (7)$$

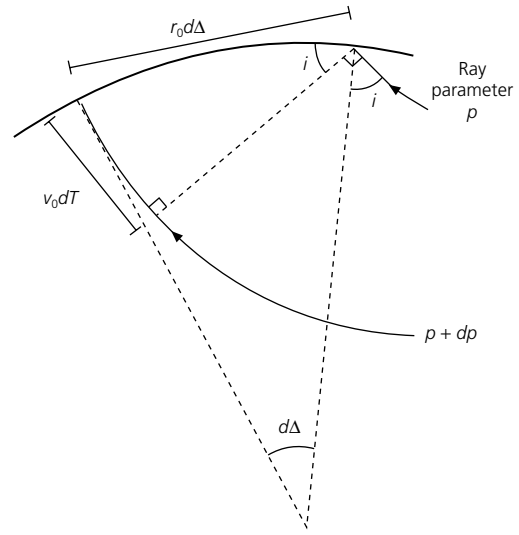


Fig. 3.4-3 Two rays with infinitesimally different ray parameters illustrating the relationship $p = dT/d\Delta$.

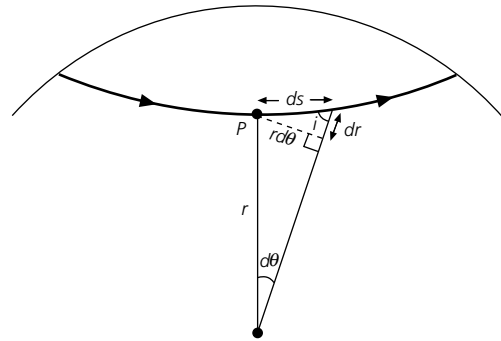


Fig. 3.4-4 Variables defining ds , a portion of the ray path subtending an angle $d\theta$.

Thus, as for the flat layer case (Section 3.3.1) the ray parameter is the reciprocal of the apparent velocity along the surface, c_x :

$$p = \frac{1}{c_x} = 1 / \left(\frac{d\Delta}{dT} \right) = \frac{dT}{d\Delta}. \quad (8)$$

Hence the ray parameter can be measured from the difference in arrival times at nearby stations. Conversely, the slope of a travel time curve $T(\Delta)$ is the ray parameter of the ray emerging at a distance Δ .

Because the geometry is spherical, it is natural to describe the ray path in polar coordinates. Consider (Fig. 3.4-4) the point P on the ray path with polar coordinates (r, θ) . A small portion of the ray path, ds , subtends an angle at the center of the earth $d\theta$, so

$$(ds)^2 = (dr)^2 + r^2(d\theta)^2 \quad \text{and} \quad \sin i = r \frac{d\theta}{ds}. \quad (9)$$

Substitution in Snell's law (Eqn 3) gives

$$p = \frac{r \sin i}{v} = \frac{r^2}{v} \frac{d\theta}{ds}. \quad (10)$$

We thus use Eqns 9 and 10 to form and equate two expressions for $(ds/d\theta)^2$,

$$\frac{r^4}{p^2 v^2} = \left(\frac{dr}{d\theta} \right)^2 + r^2, \quad (11)$$

and manipulate them to obtain

$$d\theta = \frac{\pm p dr}{r(\zeta^2 - p^2)^{1/2}}, \quad (12)$$

where ζ is defined by $\zeta = r/v$. Integrating this expression from the source depth, which we assume to be the surface r_0 , to the deepest point on the ray r_p , and doubling to account for the upward path, gives

$$\Delta(p) = \int d\theta = 2p \int_{r_p}^{r_0} \frac{dr}{r(\zeta^2 - p^2)^{1/2}}. \quad (13)$$

This integral gives the angular distance Δ traveled by the ray with ray parameter p in an earth with a velocity distribution $v(r)$.

A similar integral expression for the travel time of this ray comes from combining Eqns 9 and 10,

$$\frac{p^2 v^2}{r^2} = r^2 \left(\frac{d\theta}{ds} \right)^2 = 1 - \left(\frac{dr}{ds} \right)^2, \quad (14)$$

so that a portion of the ray path is

$$ds = \pm \frac{r}{v} \frac{dr}{(\zeta^2 - p^2)^{1/2}}. \quad (15)$$

Thus the travel time, defined by the integral of the slowness along the ray path, is given by

$$T(p) = \int \frac{ds}{v} = 2 \int_{r_p}^{r_0} \frac{\zeta^2 dr}{r(\zeta^2 - p^2)^{1/2}}. \quad (16)$$

These integral expressions for the distance $\Delta(p)$ and travel time $T(p)$ of a ray in a spherical geometry are analogous to those for $x(p)$ (Eqn 3.3.25) and $T(p)$ (Eqn 3.3.26) in a layered

material. As written, the integrals are from the surface to the bottoming depth, with the factor of 2 accounting for the return trip to the surface. If the source is not at the surface, the limits of integration are changed appropriately.

For the flat geometry, we found it useful to describe the travel time curve in terms of its slope, the ray parameter, p , and the time axis intercept of its tangent, τ (Section 3.3.2). To do the same for the spherical geometry, we write the travel time curve as

$$T(p) = p\Delta(p) + \tau(p). \quad (17)$$

We then evaluate the function

$$\tau(p) = T(p) - p\Delta(p) \quad (18)$$

using the integral expressions (Eqns 13 and 16), and find that

$$\tau(p) = 2 \int_{r_p}^{r_0} \frac{(\zeta^2 - p^2)^{1/2}}{r} dr. \quad (19)$$

This formulation can be used to invert travel time curves for velocity structure.

3.4.2 Velocity distributions

Different distributions of velocity with depth produce characteristic travel time curves. Figure 3.4-5 (overleaf) shows the usual situation in which velocity increases slowly with depth. Given two rays, the one with a smaller angle of incidence at the source has a smaller p , thus bottoms deeper at a point with smaller r_p and larger v_p , and eventually emerges further from the source. Thus the ray parameter decreases, and travel time increases, monotonically with distance, Δ . The travel time curve, $T(\Delta)$, is concave downward because its slope, $p(\Delta)$, decreases with distance ($dp/d\Delta = d^2T/d\Delta^2 < 0$). The intercept-slowness curve, $\tau(p)$, is smooth. To show these relations in different ways, the plots in Fig. 3.4-5 are aligned so that the distance axis is common to the ray path, $T(\Delta)$, and $p(\Delta)$ plots, the depth axis is the same for the ray path and velocity-depth plots, and the time axis is the same for the $T(\Delta)$ and $\tau(p)$ plots.

A more complicated situation occurs when velocity increases rapidly with depth (Fig. 3.4-6). Rays that bottom either above or below the region of high velocity gradient behave as in Fig. 3.4-5, so the corresponding portions of the travel time and ray parameter curve show T increasing with Δ , and p decreasing with Δ . By contrast, rays that bottom in the region of high velocity gradient are bent upward more and emerge at smaller values of Δ than would otherwise be the case. As a result, three rays with different ray parameters emerge at the same distance Δ . Thus the $p(\Delta)$ and $T(\Delta)$ curves have three distinct branches. On the two normal forward branches $dp/d\Delta < 0$. However, on

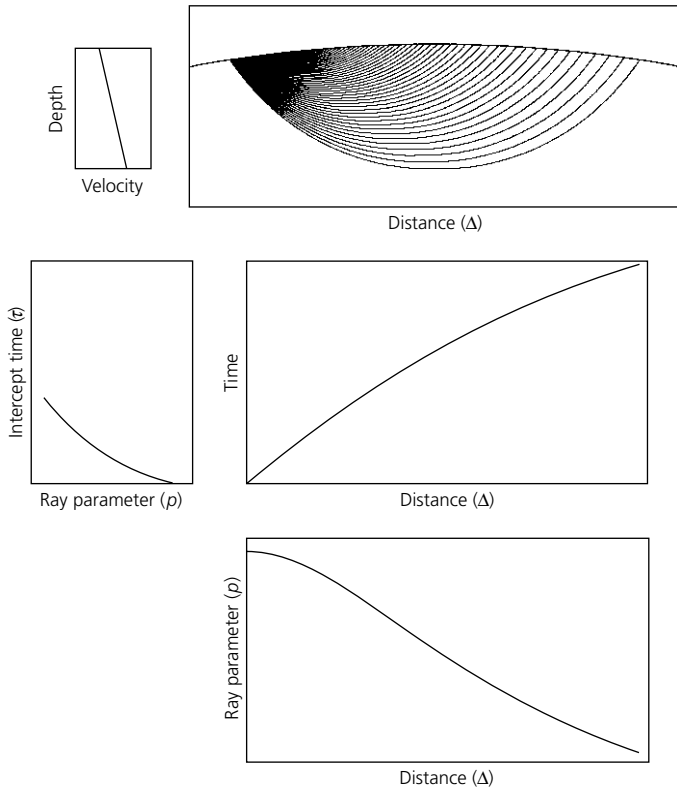


Fig. 3.4-5 Ray paths, $T(\Delta)$, $p(\Delta)$, and $\tau(p)$ relationships for velocity increasing slowly with depth.

the back branch, Δ decreases with decreasing p , so $dp/d\Delta > 0$. Thus rays with smaller incidence angles arrive closer to the source, giving a characteristic *triplication* in the travel time curve and a reversal in the $p(\Delta)$ curve. We will see in the next section that triplications are observed in the travel time curve for waves in the mantle, due to velocity increases that are thought to result from mineral phase transitions.

A triplication is similar to the travel time curves for the direct, reflected, and head waves for a layer over a halfspace (Fig. 3.2-2). The back branch of the triplication is analogous to the reflection, and the two forward branches are analogous to the direct and head waves. As the velocity increase becomes sharper and more like the sharp jump between a layer and halfspace, the back branch extends further in either direction, so the triplication looks increasingly like the travel times for a layer over a halfspace.

As we discussed in Section 2.8.4, geometric ray theory gives information about amplitudes as well as travel times. Because the rays plotted left the source at uniform increments of angle, the amplitude expected at some distance depends on *geometric spreading*, or the density of rays arriving. We expect high amplitudes where rays are concentrated, and low amplitudes where rays are sparse. Mathematically, the concentration of

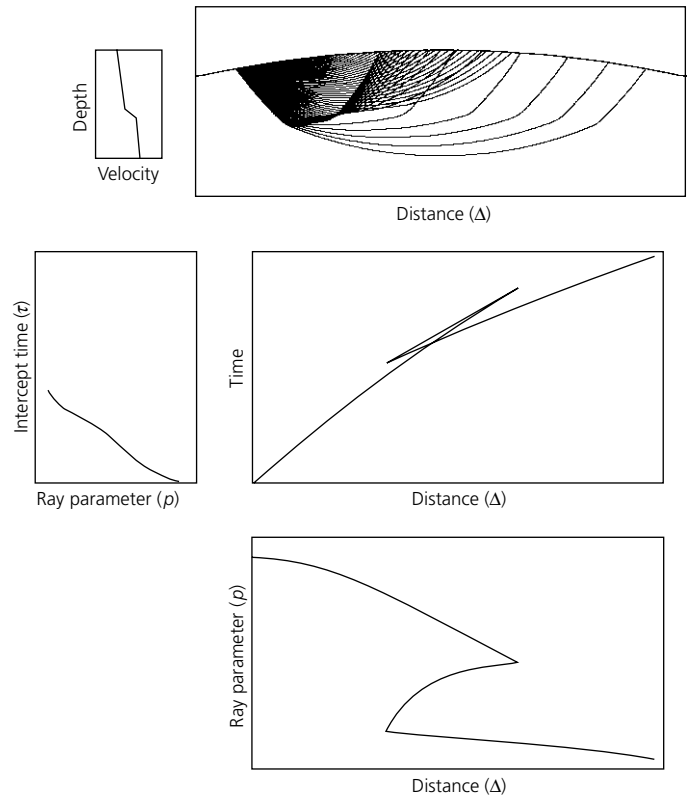


Fig. 3.4-6 A triplication occurs if velocity increases rapidly, because at some distances three rays arrive. The triplication appears as the three branches in the $T(\Delta)$ and $p(\Delta)$ curves. The cusps on the travel time curve where the branches meet correspond to reversals in the $p(\Delta)$ plot.

rays is proportional to $di/d\Delta$, the range of incidence angles for the rays that arrive in a given distance. To find this, we differentiate the definition of the ray parameter (Eqn 7),

$$\frac{d^2T}{d\Delta^2} = \frac{dp}{d\Delta} = \frac{d(r \sin i/v)}{d\Delta} = \frac{r}{v} \cos i \frac{di}{d\Delta}. \quad (20)$$

Thus the amplitude is proportional to the second derivative of the travel time curve, or the derivative of the $p(\Delta)$ curve. For a triplication, the back branch meets the two forward branches at two points on the travel time and $p(\Delta)$ curves. Here $dp/d\Delta = \infty$, so large amplitudes are expected. This situation is called a *caustic*.

A third important case is a low-velocity zone, where velocity decreases with depth and then increases (Fig. 3.4-7). Rays entering the low-velocity zone bend down, rather than up, so no rays bottom there. To see this, note that for a ray to bottom, it must turn upward (to a larger angle of incidence) as it goes deeper (to smaller values of r), so that $di/dr < 0$. Conversely, if $di/dr > 0$, the ray turns downward and cannot bottom. These conditions can be written in terms of the velocity–depth function by differentiating both sides of

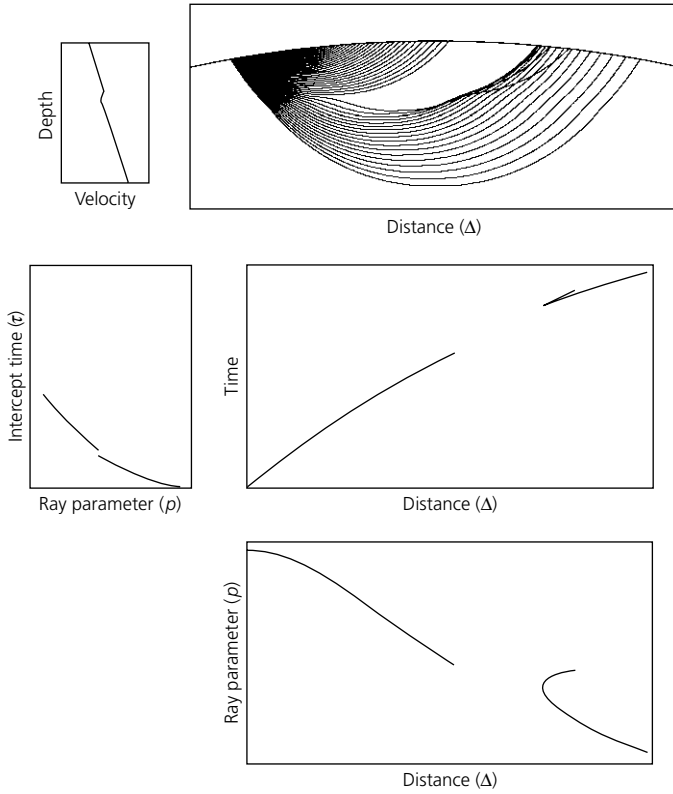


Fig. 3.4-7 A low-velocity zone gives rise to a shadow zone, a distance range where no direct geometric arrivals appear, and hence discontinuous $T(\Delta)$, $p(\Delta)$, and $\tau(p)$ curves.

$$\sin i = \frac{pv}{r}, \quad (21)$$

yielding

$$\cos i \frac{di}{dr} = p \left(\frac{1}{r} \frac{dv}{dr} - \frac{v}{r^2} \right) = \sin i \left(\frac{1}{v} \frac{dv}{dr} - \frac{1}{r} \right) \quad (22)$$

and thus

$$\frac{di}{dr} = \tan i \left(\frac{1}{v} \frac{dv}{dr} - \frac{1}{r} \right). \quad (23)$$

The condition that no rays bottom in a depth region where di/dr is positive implies that the velocity decreases fast enough that

$$\frac{dv}{dr} > \frac{v}{r}. \quad (24)$$

This situation causes a *shadow zone*, a region of the earth's surface where no rays arrive. Just below the low-velocity zone, rays reach a given Δ by two paths, giving two values of p

and travel time for a given distance. The back branch, with $dp/d\Delta > 0$, corresponds to the rays that would have bottomed at the depth of the low-velocity zone, had the velocity there been high enough. The forward branch, which continues to greater distances, corresponds to the rays that bottom deeper. The concentration of rays just past the shadow zone corresponds to the point where the two branches meet. Here $dp/d\Delta = \infty$, so large amplitudes occur. We will see that this situation occurs as a result of the drop in velocity across the core-mantle boundary, which gives rise to a shadow zone.

3.4.3 Travel time curve inversion

To infer the distribution of velocity with depth, travel time curves are compiled from seismograms recorded at different source-receiver distances. The inverse problem of deriving velocity structure from the $T(\Delta)$ curves can be done in various ways. One is to use a computer program, based on Snell's law, to trace rays through different velocity structures and compute the corresponding travel time curves. Figures 3.4-5-7 were derived this way. This approach solves the inverse problem by solving the forward problem repeatedly until a satisfactory solution is found. An alternative is to solve the inverse problem directly by deriving $v(r)$ from $T(\Delta)$.

Various methods have been used to solve the inverse problem. A classic one is the *Herglotz-Wiechert integral*. This approach is based on Eqn 13, which gives the distance traveled by a ray with ray parameter p as a function of the velocity structure

$$\Delta(p) = 2p \int_{r_p}^{r_0} \frac{dr}{r(\zeta^2 - p^2)^{1/2}}, \quad (25)$$

where $\zeta = r/v$, and p is the ray parameter for the ray arriving at Δ . This can be converted to

$$\int_0^{\Delta_1} \cosh^{-1} \left(\frac{p(\Delta)}{\zeta_1} \right) d\Delta = \pi \ln \left(\frac{r_0}{r_1} \right), \quad (26)$$

where $\zeta_1 = r_1/v_1$ at radius r_1 , the bottoming point of the ray that emerges at Δ_1 .¹ This formula is used by starting with an observed travel time curve, $T(\Delta)$, and forming its derivative $dT/d\Delta = p(\Delta)$ numerically. The integral is done numerically from $\Delta = 0$ to $\Delta = \Delta_1$, using the fact that $\zeta_1 = dT/d\Delta$ at a distance Δ_1 . The equation then gives the radius, r_1 , at which the velocity is r_1/ζ_1 .

This method sometimes fails when velocity decreases with depth, giving a low-velocity zone. In some such cases, it can still be applied using a method called "earth stripping." To do this, $v(r)$ is found down to the low-velocity zone using the Herglotz-Wiechert integral. Equations 13 and 16 are then used with r' ,

¹ Bullen and Bolt (1985).

the outer radius of the low-velocity zone, substituted for r_p , to find the distance and time the ray traveled on its way down to the low-velocity zone. Subtracting these from the known $T(\Delta)$ curve gives a $T'(\Delta)$ curve for a “mini-earth” with radius r' . Because in this “mini-earth” the velocity increases with depth, the Herglotz–Wiechert method is applied again.

3.5 Body wave travel time studies

We saw in the last section that travel time data can be used to determine seismic velocity as a function of depth. Beginning early in the 1900s, travel time tables were compiled by combining data from many earthquakes observed at various epicentral distances. These seismological observations provide the primary data for our view of the basic features of the earth’s velocity structure. This picture, an essentially layered earth composed of a thin crust, a mantle, a liquid outer core, and a solid inner core, is key to our thinking about how the earth evolved and operates. This concept was largely developed by the 1940s, as illustrated in Fig. 3.5-1, showing the classic Jeffreys–Bullen¹ (JB) earth model. The JB model treated the earth as a series of shells, characterized by the behavior of the velocity with depth (Table 5.1-1). The mantle was divided into an *upper mantle* (region B) and a *lower mantle* (region D),

Table 3.5-1 Regions in Jeffreys–Bullen earth model.

Region	Depth (km)	Features of region
A	33	Crustal layers
B	413	Upper mantle: steady positive P and S velocity gradients
C	984	Mantle transition region
D	2898	Lower mantle: steady positive P and S velocity gradients
E	4982	Outer core: steady positive P velocity gradient
F	5121	Core transition: negative P velocity gradient
G	6371	Inner core: small positive P velocity gradient

Source: After Bullen and Bolt (1985).

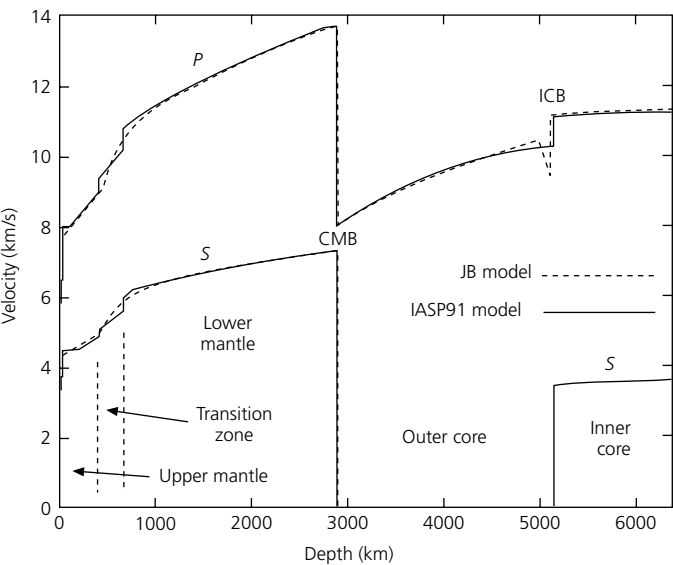


Fig. 3.5-1 Comparison of the classic Jeffreys–Bullen earth model (Jeffreys and Bullen, 1940) and a newer model, IASP91 (Kennett and Engdahl, 1991). Although IASP91 and its successor, AK135 (Kennett *et al.*, 1995), have improved resolution in the mantle transition zone and the core, the newer models are generally similar to that derived using hand-cranked calculators.

¹ The model was derived from extensive joint research into earth structure by Sir Harold Jeffreys (1891–1989), who established in 1926 that the core was liquid, and Keith Bullen (1906–76).

both of which had smooth velocity gradients. These upper and lower mantle regions were separated by region C, the *mantle transition zone* where velocities increase rapidly with depth. Below the *core–mantle boundary* (CMB), the core was divided into an *outer core* (region E) and an *inner core* (region G), separated by a transition zone (region F). The *inner core boundary* (ICB) separated regions F and G. Subsequently, the lower mantle was divided into regions D’ (1000–2700 km depth), most of the lower mantle with a smooth velocity gradient, and D’’ (2700–2900 km), the zone above the core–mantle boundary with a reduced velocity gradient.

Subsequent studies have derived models, such as the IASP91 model, also shown in Fig. 3.5-1, which confirm the basic structure of the JB model and provide better resolution of important regions. For example, the JB model did not resolve shear velocities in the inner core, whereas recent models have finite S velocity in the inner core, implying that it is solid. Similarly, recent models provide more details about the mantle transition zone and the core–mantle boundary, and do not include the velocity “notch” at the inner core–outer core boundary.

Jeffreys’ and Bullen’s derivation of a radially symmetric earth model from travel time observations converted the previous crude picture of the earth into one that has since changed only in detail. More recent radial velocity models do not differ much from each other, so they are likely to be converging on an accurate radial model for the earth. Such average, or *reference*, models and travel time curves, such as JB, IASP91, and PREM (for Preliminary Reference Earth Model, Section 3.8), are derived from data around the world and so average over local variations in structure. Regional differences can then be viewed as perturbations relative to a reference model.

However, lateral differences in structure can be significant and provide insight into tectonic processes. Thus a major current goal of seismology is to define the three-dimensional velocity structure that results from the fact that the earth is a